Article

Comparative Electrochemical Impedance Spectroscopy Analysis of Monocrystalline, Polycrystalline, and Amorphous Silicon Solar Cells

Kevin Morkar 1,*

1 Department of Engineering, De Montfort University, Leicester, United Kingdom; kevinmorkar@gmail.com

- Correspondence: kevinmorkar@gmail.com; Tel.: +91-8799608668

Abstract

This study presents a comparative analysis of three photovoltaic (PV) cell technologies — monocrystalline silicon (c-Si), polycrystalline silicon (poly-Si), and amorphous silicon (a-Si) — using electrochemical impedance spectroscopy (EIS), current-voltage (I-V), and capacitance-voltage (C-V) characterisation techniques. **Methods:** Cells were evaluated under illumination (833 W/m²), dark conditions with external applied voltage, and across a frequency range of 20 Hz to 1 MHz using a Keysight Agilent impedance analyser. Key parameters including fill factor, conversion efficiency, ideality factor, reverse saturation current, barrier height, charge transfer resistance, and junction capacitance were systematically extracted and compared. **Results:** Monocrystalline silicon demonstrated the most favourable impedance characteristics with a well-defined Nyquist semicircle and capacitance of 3.3 nF/cm², while polycrystalline silicon achieved the highest optical conversion efficiency of 5.62%. Amorphous silicon exhibited anomalous high-frequency behaviour and critically degraded C-V characteristics, operating reliably only below 10 kHz, suggesting its preferential suitability for low-intensity indoor applications. **Conclusions:** These findings contribute to the broader understanding of defect characterisation and performance benchmarking of silicon PV technologies using EIS as a diagnostic tool.

Keywords: Electrochemical Impedance Spectroscopy; Solar Cells; Nyquist Plot; Photovoltaics; Silicon; Fill Factor; Capacitance-Voltage; PV Characterisation

---

1. Introduction

Global energy demand continues to rise, with fossil fuel dependence driving greenhouse gas emissions and accelerating climate change. Photovoltaic (PV) technology offers a scalable, zero-emission alternative, converting sunlight directly into electricity with no moving parts and minimal operating costs [1]. Since Becquerel's discovery of the photovoltaic effect in 1839 and the development of the first functional silicon solar cell at Bell Labs in 1954 [2,3], photovoltaics have evolved through three generations — from crystalline silicon to amorphous thin films and emerging perovskite technologies [4].

Despite efficiency improvements, the characterisation of internal electrical behaviour — particularly recombination mechanisms, junction quality, and degradation — remains critical for optimising cell performance. Electrochemical Impedance Spectroscopy (EIS) is a powerful non-destructive technique that applies a small AC perturbation across a range of frequencies and analyses the resulting impedance response. By fitting the data to an equivalent circuit model, EIS enables extraction of key parameters including charge transfer resistance, double-layer capacitance, series resistance, and shunt resistance [5,6].

While EIS has been widely applied to perovskite and dye-sensitised solar cells, comparative impedance analysis of conventional silicon cell types — monocrystalline, polycrystalline, and amorphous — remains underexplored as a unified benchmarking methodology. This study addresses that gap by performing systematic multi-technique characterisation across all three silicon PV technologies and establishing a figure of merit based on impedance behaviour.

---

2. Materials and Methods

2.1. Solar Cells Tested

Three commercial silicon solar cells were characterised:

- **Monocrystalline silicon (c-Si):** 2 cm × 1.5 cm, 4-cell configuration
- **Polycrystalline silicon (poly-Si):** 4 cm × 4 cm, 4-cell configuration
- **Amorphous silicon (a-Si):** 5 cm × 4.6 cm

2.2. Equivalent Circuit Model

A solar cell can be modelled as a p-n junction diode in parallel with a current source, shunt resistance (R_{sh}), and series resistance (R_s). The impedance of this circuit is expressed as:

$$Z(\omega) = R_s + \frac{R_{sh}}{1 + j\omega C \cdot R_{sh}}$$

where ω is the angular frequency and C is the junction capacitance. The real and imaginary parts of this impedance generate the characteristic semicircular Nyquist plot, where the diameter corresponds to R_{sh} and the high-frequency intercept gives R_s [5].

2.3. Illumination Testing

I-V and P-V characteristics were measured using a **Wavelabs SINUS-70** solar simulator at an illumination intensity of 833 W/m² (350–1200 nm), at T = 293.15 K. Parameters including I_{sc}, V_{oc}, V_{mp}, I_{mp}, fill factor (FF = $P_{max}/(V_{oc} \times I_{sc})$), series resistance, shunt resistance, and efficiency ($\eta = P_{max}/P_{in} \times 100\%$) were extracted.

2.4. Dark Characterisation

Dark I-V measurements were performed inside a **Faraday cage** using an Agilent voltage source, sweeping from −2V to +2V to eliminate photogenerated noise. Ideality factor (n) and reverse saturation current were extracted via linear fitting of the log(I)-V characteristic. Ideality factor represents diode quality; ideally n = 1, with higher values indicating recombination-dominated current.

C-V measurements were performed under reverse bias to determine junction capacitance and barrier height (ϕ_b) from $1/C^2$ vs V plots, representing the built-in junction potential.

2.5. Impedance Spectroscopy

EIS measurements were performed using a **Keysight Agilent impedance analyser** with **Agilent VEE Pro** software, sweeping from 20 Hz to 1 MHz with stepped frequency intervals. Real (Z') and imaginary (Z'') impedance components were recorded and plotted as Nyquist and Bode plots. Curve fitting was applied to extract capacitance, Z_{min}, and angular frequency at Z_{min}.

---

3. Results

3.1. Illumination Characteristics

Table 1 presents the solar cell performance parameters under illumination. Polycrystalline silicon achieved the highest optical efficiency (5.62%), driven by its larger active area and higher short-circuit current. Monocrystalline silicon demonstrated a significantly superior fill factor (0.727) and lower series resistance (1.9 Ω), indicating better carrier transport. Amorphous silicon exhibited very low efficiency (0.42%) and high series resistance (36.5 Ω), consistent with its disordered structural properties and the Staebler-Wronski light-induced degradation mechanism [7].

Table 1: Solar cell performance under illumination (833 W/m²)

Parameter	Poly-Si	Mono-Si	Amorphous Si
I_{sc} (mA)	64.6	35.9	4.49
V_{oc} (V)	2.41	2.50	2.44
Fill Factor	0.481	0.727	0.739
Efficiency (%)	5.62	2.60	0.42
R_{series} (Ω)	13.3	1.9	36.5
R_{shunt} (kΩ)	1.3	4.72	0.05

3.2. Dark I-V and Ideality Factor

Table 2 presents dark characterisation parameters. The ideality factors for all three cells exceeded the ideal value of 1, indicating the presence of recombination-dominated current transport rather than pure diffusion. This is typical of multi-cell series configurations. Notably, amorphous silicon achieved the lowest ideality factor (7.56) and nanoscale leakage current despite its poor illumination performance — suggesting that its junction quality is structurally sound but its optical conversion is limited by carrier mobility constraints inherent to amorphous material.

Table 2: Dark characterisation parameters

Parameter	**Poly-Si**	**Mono-Si**	**Amorphous Si**
Ideality Factor (n)	10.73	9.63	7.56
Leakage Current (I_0)	High	~nA	~nA

3.3. C-V Analysis and Barrier Height

Polycrystalline silicon exhibited a linear C-V characteristic with a calculated barrier height of 0.697 eV — consistent with standard silicon p-n junction values. Monocrystalline silicon yielded a barrier height of 0.50 eV, also within expected ranges, confirming functional junction integrity under reverse bias.

Amorphous silicon demonstrated critically abnormal C-V behaviour at both 1000 kHz and 100 kHz, with an unquantifiable barrier height suggesting junction breakdown at high frequencies. At a reduced frequency of 10 kHz, partial linearity was recovered, though the calculated barrier height remained anomalously high at approximately 203 eV — indicative of a degraded or structurally compromised junction. This frequency-dependent anomaly is attributed to the high defect density ($\sim 10^{16}/cm^3$) and disordered bond structure of amorphous silicon [7].

3.4. Impedance Spectroscopy Analysis

Table 3 presents impedance spectroscopy results. Monocrystalline silicon produced the most well-defined Nyquist semicircle with a high shunt resistance intercept of 832 kΩ, indicating minimal leakage pathways and strong diode behaviour. Its capacitance of 3.3 nF/cm² — though lower than typical reference values — reflects a high-quality junction with low recombination losses.

Polycrystalline silicon produced a good but incomplete semicircle (extrapolated intercept at 147.5 kΩ), with capacitance of 37.8 nF/cm² — a reasonable value consistent with its structural grain boundary characteristics introducing additional capacitive contributions.

Amorphous silicon failed to produce a recognisable impedance semicircle under standard conditions. Two partial curves were identified through graphical decomposition, yielding capacitances of 10.2 nF/cm² and 365 nF/cm². The large disparity between these values, combined with its anomalous C-V behaviour, strongly suggests a bifurcated or defect-compromised junction topology — consistent with the high dangling bond density and structural disorder characteristic of a-Si:H material [7,8].

Table 3: Impedance spectroscopy results

Parameter	Poly-Si	Mono-Si	Amorphous Si
Nyquist Semicircle	Partial	Well-defined	Absent (dual partial)
Capacitance (nF/cm²)	37.8	3.3	10.2 / 365
Z_{min} (kΩ)	0.65	2.36	0.025
X-axis intercept (kΩ)	147.5	832	—

---

4. Discussion

This study demonstrates that EIS, in conjunction with I-V and C-V characterisation, provides a comprehensive diagnostic framework for benchmarking silicon solar cell quality. The results can be interpreted from several perspectives:

Junction Quality: Monocrystalline silicon exhibits superior junction quality with minimal defect states, as evidenced by its well-defined Nyquist semicircle and high shunt resistance. This makes it the most reliable candidate for stable long-term photovoltaic operation in grid-connected applications.

Optical Efficiency vs Electrical Quality: While polycrystalline silicon achieved the highest optical conversion efficiency (5.62%), its elevated ideality factor and lower shunt resistance suggest potential susceptibility to early-stage junction degradation under prolonged stress. This highlights the distinction between short-term optical performance and long-term electrical reliability.

Material-Specific Limitations: The anomalous high-frequency behaviour of amorphous silicon confirms fundamental limitations imposed by its disordered structure. The Staebler-Wronski effect and high dangling bond density restrict its reliable operation to frequencies below 10 kHz, making it preferentially suitable for low-intensity indoor photovoltaic applications rather than outdoor high-frequency grid-connected systems.

These findings align with previous studies on silicon-based PV technologies [1,6] and extend the application of EIS as a quality control tool to conventional silicon cell types, complementing its established use in emerging perovskite and dye-sensitised solar cell research [9].

---

5. Conclusions

This study demonstrates that electrochemical impedance spectroscopy, combined with current-voltage and capacitance-voltage characterisation, provides a powerful diagnostic framework for silicon solar cell quality assessment. Three key conclusions emerge:

1. **Monocrystalline silicon** exhibits the best overall junction quality — well-defined Nyquist semicircle, low leakage current, and high shunt resistance — making it the most reliable candidate for stable long-term photovoltaic operation.

2. **Polycrystalline silicon** achieves the highest optical conversion efficiency (5.62%) under the test conditions, though its elevated ideality factor and leakage current indicate potential susceptibility to junction stress.

3. **Amorphous silicon** demonstrates poor high-frequency impedance and C-V behaviour, with reliable operation limited to below 10 kHz — confirming its preferential suitability for low-intensity indoor applications rather than outdoor high-frequency grid-connected systems.

These findings reinforce EIS as a valuable non-destructive tool for PV cell classification and quality control, with direct implications for the characterisation of next-generation solar technologies including perovskite and tandem junction cells.

Author Contributions: Conceptualisation, K.M.; methodology, K.M.; formal analysis, K.M.; investigation, K.M.; data curation, K.M.; writing—original draft preparation, K.M.; writing—review and editing, K.M. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Acknowledgments: The author acknowledges De Montfort University for providing laboratory facilities and equipment access for this research. During the preparation of this manuscript, the author used Perplexity AI for the purposes of manuscript formatting and reference organization. The author has reviewed and edited the output and takes full responsibility for the content of this publication.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

EIS — Electrochemical Impedance Spectroscopy
PV — Photovoltaic
c-Si — Monocrystalline Silicon
poly-Si — Polycrystalline Silicon
a-Si — Amorphous Silicon
I-V — Current-Voltage
C-V — Capacitance-Voltage
FF — Fill Factor

References

1. Guerra, N.; Leitão, J.; Guimarães, L.; Jesus, I. Operation and physics of photovoltaic solar cells: an overview. *I+D Tecnológico* **2018**, *14*, 84–95.

2. Aps.org. This Month in Physics History: April 25, 1954 – Bell Labs Demonstrates the First Practical Silicon Solar Cell. American Physical Society, **2022**.

3. Binz, C.; Gosens, J.; Hansen, T.; Hansen, U. Spatial lifecycles of cleantech industries – The global development history of solar photovoltaics. *Energy Policy* **2017**, *101*, 386–402.

4. Yan, J.; Saunders, B. Third-generation solar cells: a review and comparison of polymer:fullerene, hybrid polymer and perovskite solar cells. *RSC Advances* **2014**, *4*, 43286–43314.

5. Bisquert, J.; Garcia-Belmonte, G.; Fabregat-Santiago, F.; Ferriols, N.; Bogdanoff, P.; Pereira, E. Doubling exponent models for the analysis of porous film electrodes by impedance. Relaxation of TiO2 nanoporous in aqueous solution. *Journal of Physical Chemistry B* **2000**, *104*, 2287–2298.

6. Suresh, M. Measurement of solar cell parameters using impedance spectroscopy. *Solar Energy Materials and Solar Cells* **1996**, *43*, 21–28.

7. Staebler, D.L.; Wronski, C.R. Reversible conductivity changes in discharge-produced amorphous Si. *Applied Physics Letters* **1977**, *31*, 292.

8. Stutzmann, M. Weak bond-dangling bond conversion in amorphous silicon. *Philosophical Magazine B* **1986**, *56*, 63–70.

9. von Hauff, E.; Klotz, D. Impedance spectroscopy for perovskite solar cells: characterisation, analysis, and diagnosis. *Journal of Materials Chemistry C* **2022**, *10*, 742–761.